Assignment 2 – Python & Linux Basics

Course: Tutedude DevOps
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Task 1: Grade Checker

1. **Objective:**

   Create a Python program that takes a score as input and prints the corresponding grade using if-else statements.

2. **Linux Commands Used:**

- mkdir tutedude_assignment2
- cd tutedude_assignment2
- touch grade_checker.py
- nano grade_checker.py
- python3 grade_checker.py

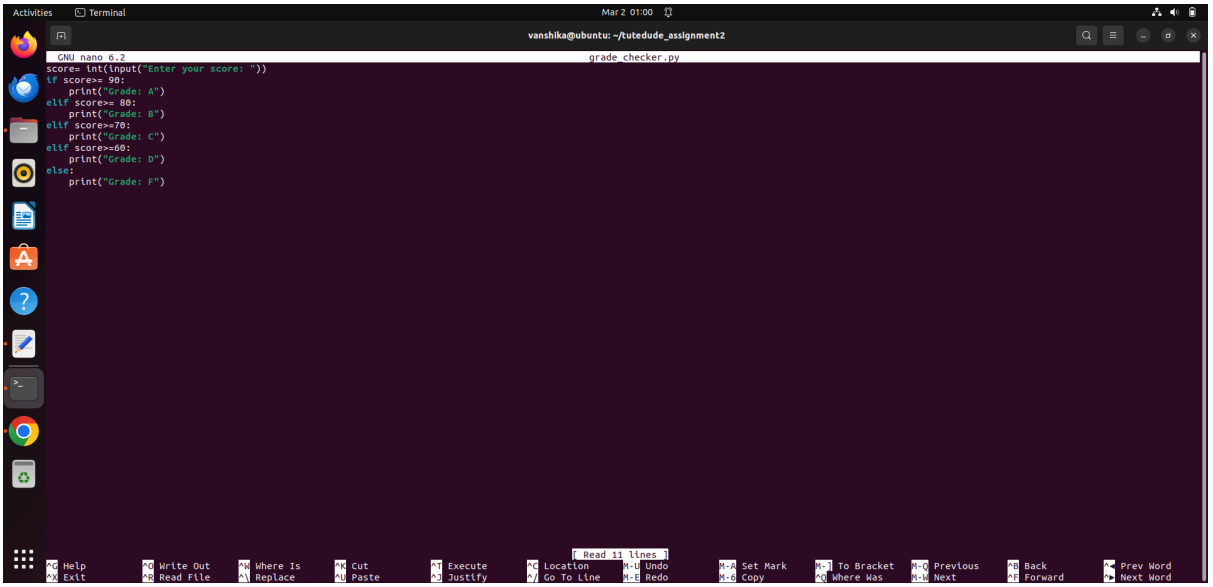
3. **Python Code:**

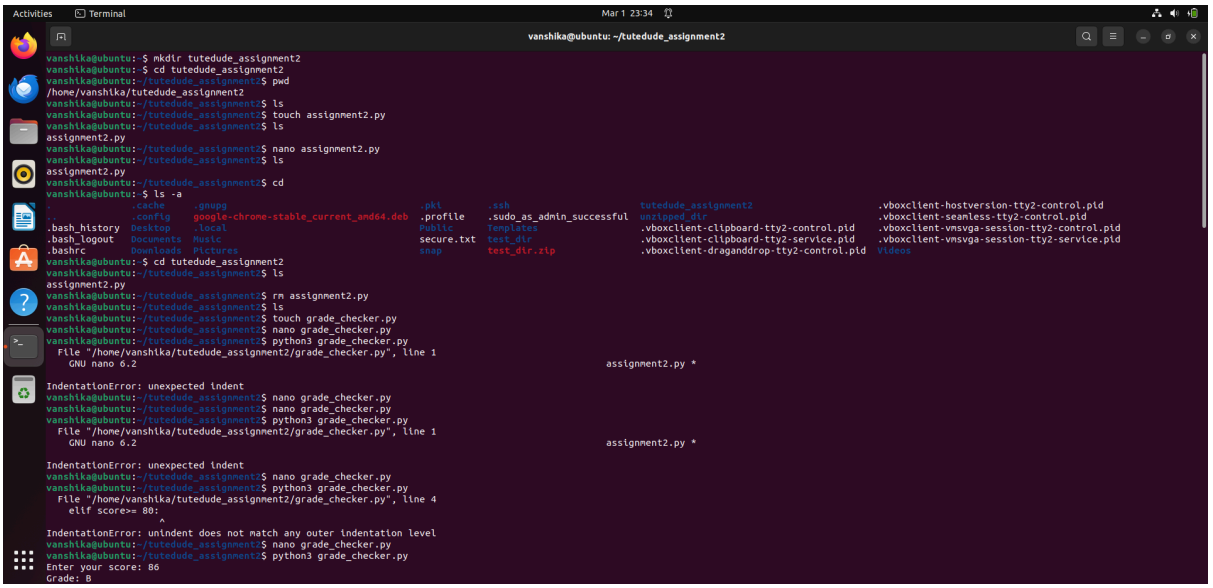
```
score = int(input("Enter your score: "))

if score >= 90:
  print("Grade: A")
elif score >= 80:
  print("Grade: B")
elif score >= 70:
  print("Grade: C")
elif score >= 60:
  print("Grade: D")
else:
  print("Grade: F")
```

4. **Explanation:**

- The program takes input from the user.

- Uses if-elif-else conditions to check score range.
- Prints the grade accordingly.





Task 2: Student Grades Dictionary

1. **Objective:**

   Create a dictionary to:

- Add a new student
- Update student grade
- Print all students

2. **Linux Commands Used:**

- touch student_grades.py
- nano student_grades.py
- python3 student_grades.py

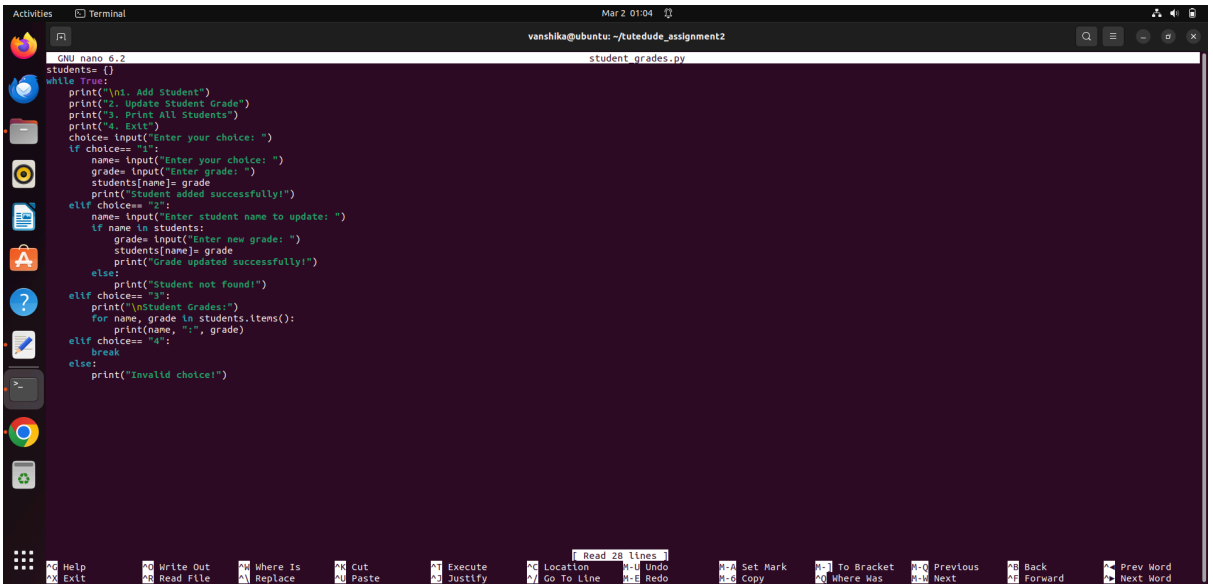
3. **Python Code:**

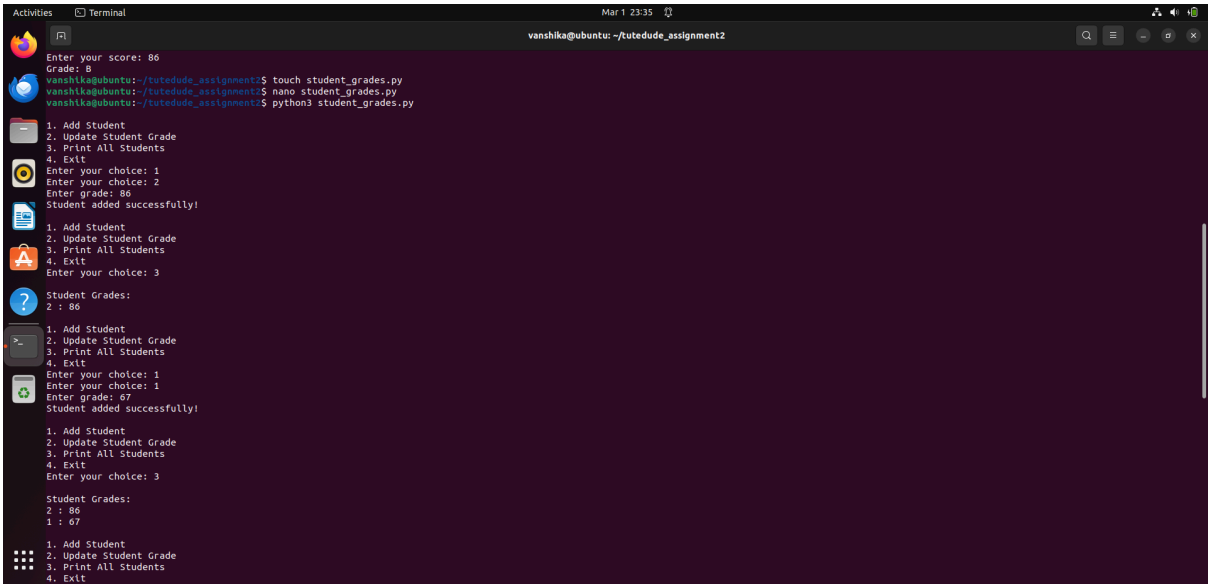
```
students = {}

while True:
    print("\n1. Add Student")
    print("2. Update Student Grade")
    print("3. Print All Students")
    print("4. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        name = input("Enter student name: ")
        grade = input("Enter grade: ")
        students[name] = grade
        print("Student added successfully!")

    elif choice == "2":
        name = input("Enter student name to update: ")
        if name in students:
            grade = input("Enter new grade: ")
            students[name] = grade
            print("Grade updated successfully!")
        else:
            print("Student not found!")

    elif choice == "3":
        print("\nStudent Grades:")
        for name, grade in students.items():
            print(name, ":", grade)
```

```
elif choice == "4":
    break

else:
    print("Invalid choice!")
```

4. **<u>Explanation:</u>**

- A dictionary named students is created.
- User chooses options from the menu.
- Dictionary operations like add and update are performed.
- Loop runs until the user exits.





Task 3: Write to a File

1. **Objective:**

- Create a text file and write content into it using file handling.

2. **Linux Commands Used:**

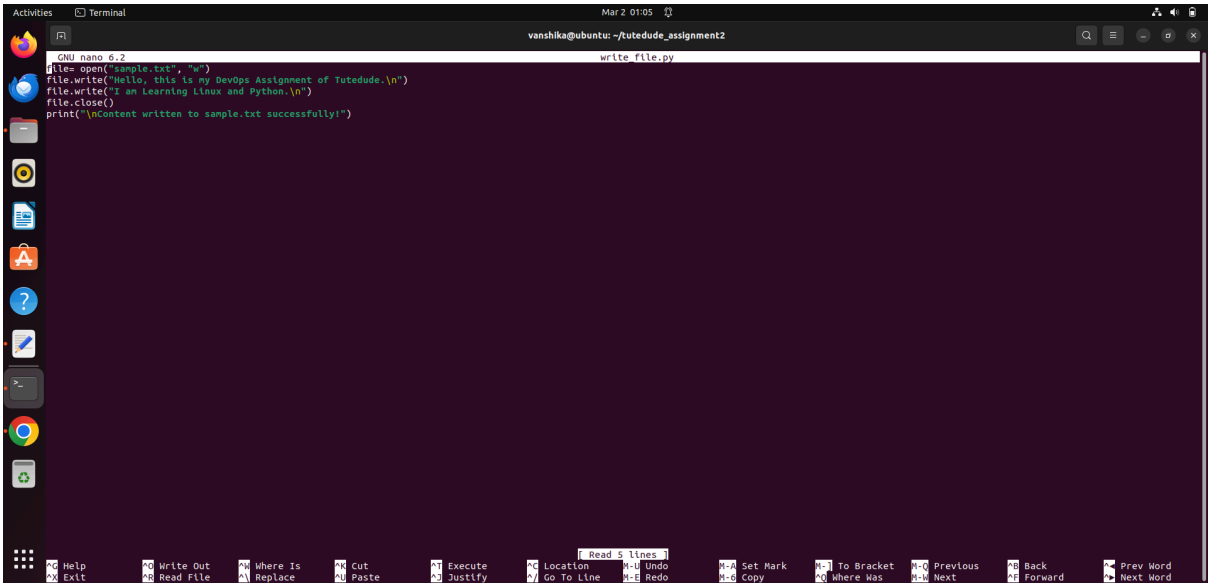
- touch write_file.py
- nano write_file.py
- python3 write_file.py
- ls
- cat sample.txt

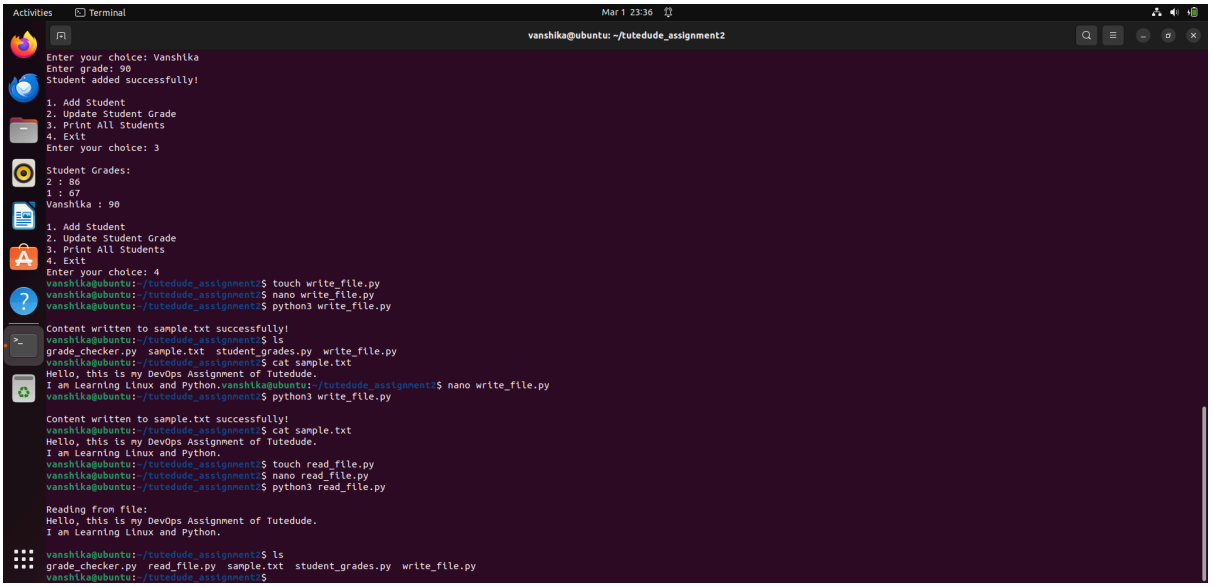
3. **Python Code:**

```
file = open("sample.txt", "w")
file.write("Hello, this is my DevOps assignment.\n")
file.write("Learning Linux and Python file handling.")
file.close()

print("Content written successfully!")
```

4. **Explanation:**

- File opened in write mode ("w").
- Content written using the write() function.
- File closed after writing.
- The sample.txt file is created in the directory.





Task 4: Read from a File

1. **Objective:**

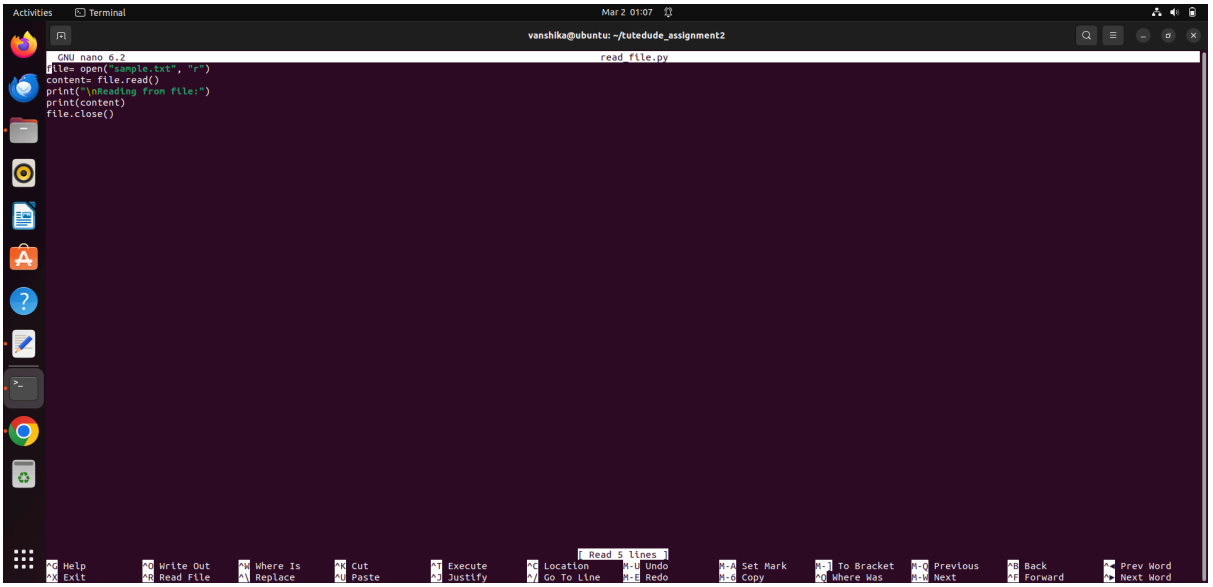
- Read content from an existing file and display it.

2. **Linux Commands Used:**

- touch read_file.py
- nano read_file.py
- python3 read_file.py

3. **Python Code:**

```
file = open("sample.txt", "r")
content = file.read()
print("File Content:\n")
print(content)
file.close()
```

4. **<u>Explanation:</u>**

- File opened in read mode ("r").
- Content stored in a variable using read().
- Printed on screen.